

Assignment – Java Web Application Development

Module: Object-Oriented Programming / Web Application Development

Level: Year 2 Undergraduate

Assignment Description

Students are required to design and develop a dynamic web application using Java technologies. The project should demonstrate understanding of server-side programming, database integration, form handling, object-oriented concepts, and web application architecture.

What Is Expected from Students

- Develop a functional Java-based web application using Servlets and JSP (or equivalent Java web technology).
- Implement user input handling and server-side validation.
- Integrate a relational database (e.g., MySQL) for data storage and retrieval.
- Apply proper MVC (Model-View-Controller) design principles.
- Demonstrate use of object-oriented programming concepts (classes, inheritance, encapsulation).
- Ensure proper session management where required.
- Design a clean and user-friendly front-end interface.
- Handle exceptions and errors appropriately.
- Organize project structure according to standard Java web application format.

Technical Requirements

- Use Java (Servlets/JSP or approved framework).
- Use a relational database for backend storage.
- Deploy and test using a web server such as Apache Tomcat.
- Follow proper package structure and naming conventions.
- Use prepared statements to prevent SQL injection.

Rules and Submission Guidelines

- The project must be developed individually unless otherwise specified.
- Copying complete projects from the internet is strictly prohibited.
- Code must be well-commented and properly formatted.
- Submit source code files along with database script file.
- Include a short project report explaining system design and functionality.
- Ensure the application runs correctly before submission.

Evaluation Criteria

- Correct implementation of Java web technologies.
- Database integration and query handling.
- Application functionality and logic.
- Code quality and structure.
- User interface design and usability.
- Documentation and overall presentation.